Features

Our primary persona is **Emily Carter**, a graduate student who frequently visits PFT to attend classes, meet faculty members, and use labs. Based on the task scenario we had developed in Milestone 1, Emily often needs to quickly locate classrooms or offices inside the building. She needs to navigate to the desired place efficiently. The following tasks and features were designed to support her needs.

Task 1: Search for a Destination: Emily needs to quickly find a specific room or office inside PFT. Since she frequently visits the same locations such as classrooms, laboratories, and professor offices, the system must allow her to easily search and retrieve destinations.

- **Feature 1- Search Bar:** The system includes a search bar on the main page. In that page, users can enter keywords such as room numbers, professor names, or class identifiers. Once the user enters a query, the system highlights possible destinations on the indoor map and displays the selected location. Generally, students often know the room number or instructor name. But sometimes she gets confused with the building layout. A flexible search function allows users to quickly identify the location of their destination and begin navigation.
- **Feature 2- Location History:** The system stores previously searched destinations. And it also displays them in a drop-down suggestion list when the user begins typing in the search bar. For example, if Emily previously searched for “Room 1240,” the location will appear in the history suggestions. Students often visit the same rooms multiple times each week. Location history helps to reduce the time required to re-enter the same destination information. Moreover, this feature improves efficiency for frequent users.

Task 2: Navigate to the Destination: Once Emily selects a destination, she needs assistance navigating through the building’s complex hallways and floors. The system provides map-based guidance and navigation instructions.

- **Feature 1- Indoor Map with Route Visualization:** The system displays an interactive indoor map of PFT showing the user’s location and the route to the selected destination. The route is highlighted to visually guide the user through hallways and intersections. Participants in the formative study reported confusion due to similar-looking corridors and unclear layout structures. Route visualization helps users understand the spatial structure of the building and reduces navigation errors.
- **Feature 2: Step-by-Step Navigation Instructions:** The system provides simple step-by-step instructions such as: (1) Walk straight 20 meters, (2) turn right at the hallway, (3) take the stairs to Floor 2. These instructions update as the user walks along the route. Static maps alone may not provide enough guidance at some decision points such as hallway intersections. In that case, step-by-step instructions help users confidently navigate through the building and reduce uncertainty.

2. Low-Fidelity Prototype

Prototype Tool: The low-fidelity prototype was created using **Figma**, a collaborative interface design tool. Figma allows rapid development of wireframes and makes it easy for team members to collaborate on interface design. The basic functionalities and the flow of our system has been shown in Figure 1.

Prototype Overview: The prototype demonstrates how users complete the two main tasks:

1. Searching for a destination
2. Navigating to the destination

The prototype consists of several screens that illustrate the user interaction flow.

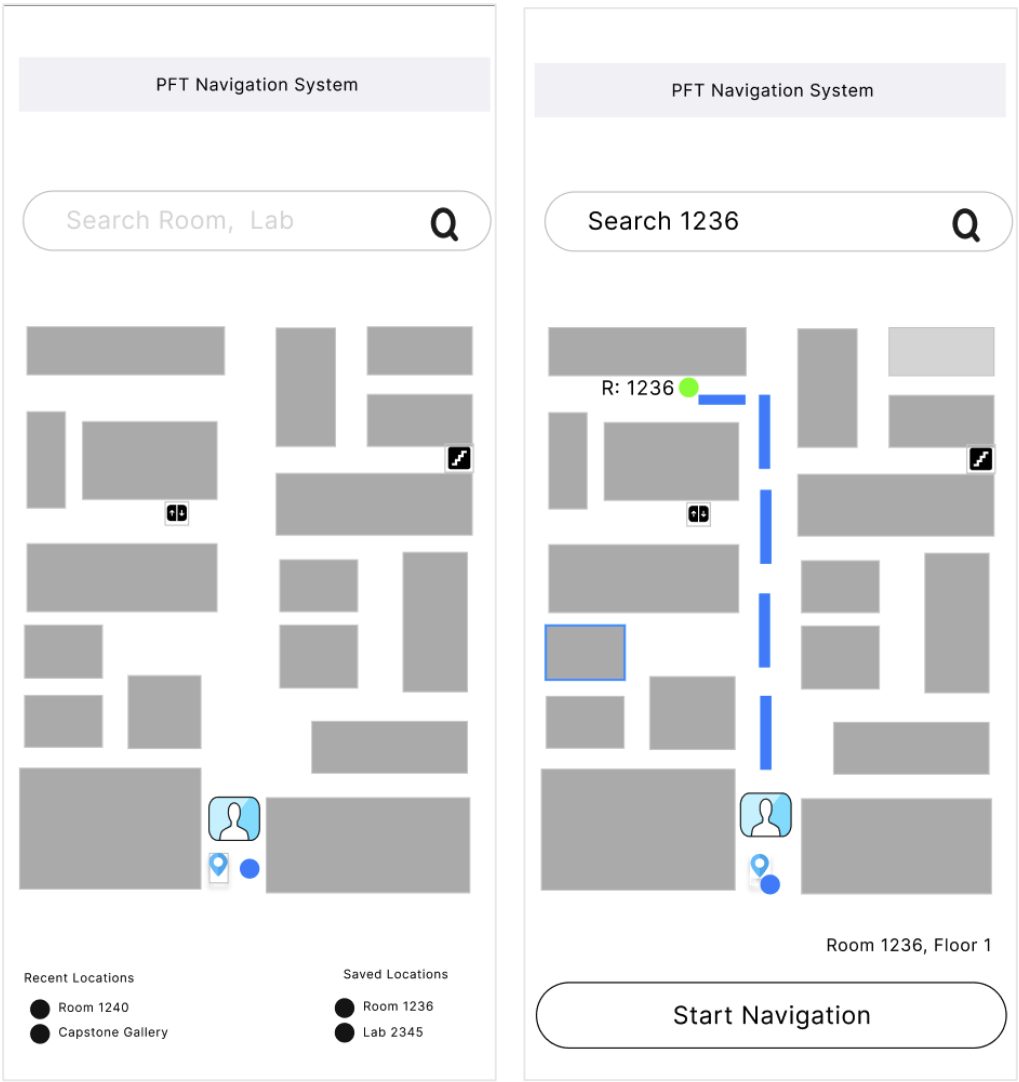

Figure 1 : a. Home Screen (left), b. Search Result Screen (right)

Screen 1- Home Screen: The home screen displays a search bar where users can enter the name or number of a destination. It also displays previously searched locations and saved tagged locations. The interaction it supports are:

- Enter a room number or keyword
- Select a previously searched location/pinned locations

This screen supports the **task of searching for a destination** quickly.

Screen 2- Search Results: After entering a search query, the system highlights the matching destination on the indoor map and displays the location details. Users can then choose to start navigation. The interaction it supports are:

- Select a destination
- Start navigation

This screen confirms that the user has selected the correct destination.

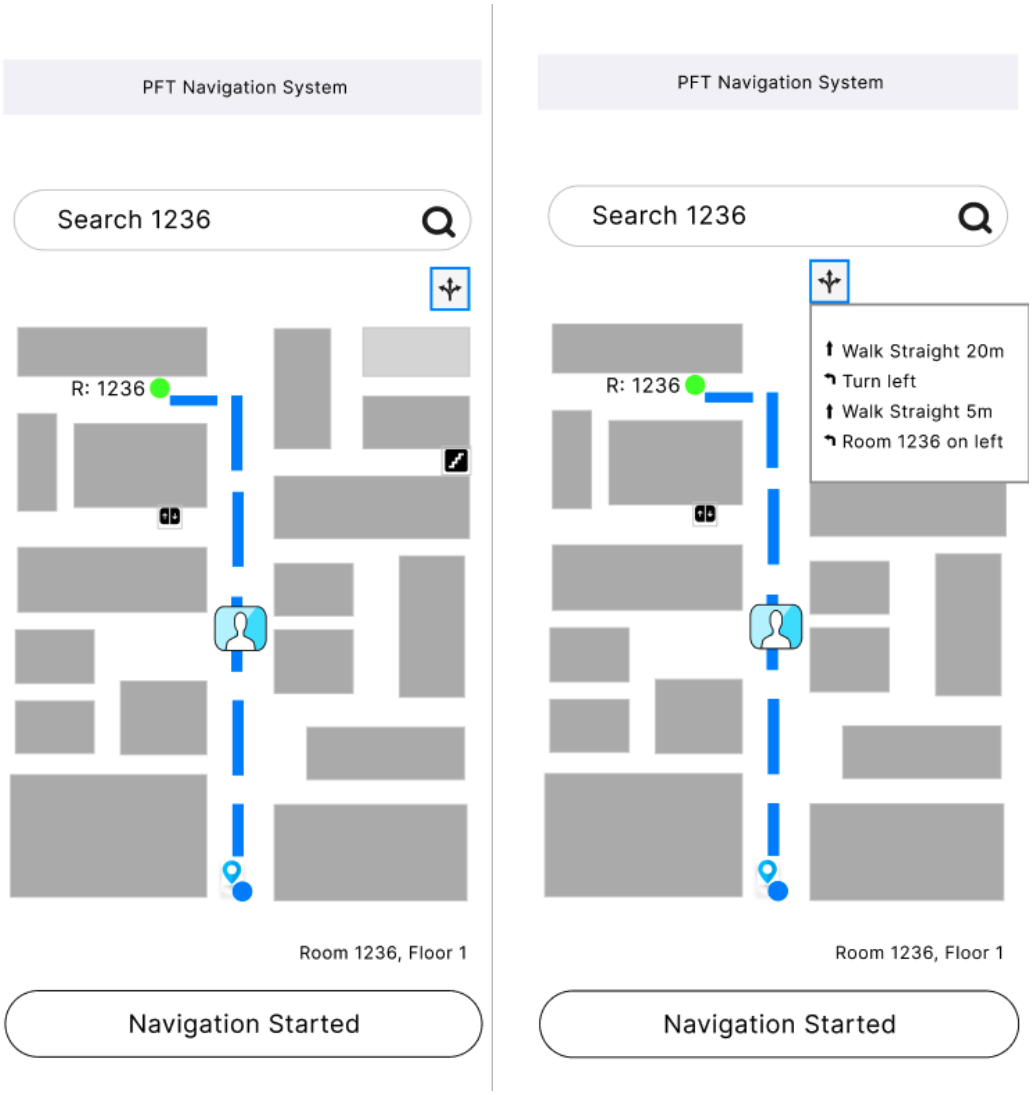

Figure 2: c. Navigation Map (left) d. Step-by-Step Navigation (right)

Screen 3- Navigation Map: This display provides a high-level spatial view of Patrick F. Taylor Hall (PFT) helps users understand their current location and the path to their desired destination. The user profile marker indicates the current location of the user, and the recommended route is shown as a highlighted route which directs the user towards the designated room or office. Multi-floor awareness is also implemented in the mapping interface as it shows the current floor level and highlights the floor transition points, such as stairs or elevators, that are required in the itinerary. The main landmarks such as elevators, stairwells, the main atrium, and vending machines can be depicted by straightforward icons so that it can be easy to navigate through the corridors that display a lot of visual similarity.

Supported interaction:

- View route
- Follow highlighted path

This screen supports a spatial understanding of the building layout.

Screen 4- Step-by-Step Navigation: The interface provides real-time guidance via a series of step-by-step instructions that can be followed during the walk. This view is also optimized to support decision points, unlike a static map overview, like the intersections of hallways, choices of stairs or elevators, and transitioning between floors. The system dynamically updates the next instruction according to the position of the user as he advances; the next turn is shown immediately after one reaches an intersection. Cursory clues can also be added to the screen that combine signage with landmarks i.e., Turn left at the elevators or walk straight 20m. These indications assist users in establishing that they are in the right direction.

Example instructions:

- Walk straight 20 meters
- Turn left at hallway intersection
- Take elevator to Floor 2

This feature helps users navigate efficiently.

Screen 5- Destination Confirmation: When the system reaches the specified endpoint, it switches to an interface where an arrival confirmation can be obtained to clearly show the end of navigation. At the top of the page, there is a conspicuous headline, like Destination Reached, and there are identifiers of the location (e.g., Room 1236 and, possibly, floor or area). The indoor map continues to display, and users can confirm their location at the desired location graphically. The accentuation of Room 1236 is used in this setup, by outlining or pinning, and the latest positional information of the user is displayed to strengthen confidence. A brief confirmation message such as You have reached Room #1236

provides another form of protection, especially relevant in cases where two or more doors or similar room names are close to each other. The interface can optionally include a simple solicitation, such as Was this route helpful? using thumbs-up/thumbs-down or star-rating systems, to collect a large volume of feedback, which increases the accuracy of routing and general usability, without interfering with the main purpose of the user.

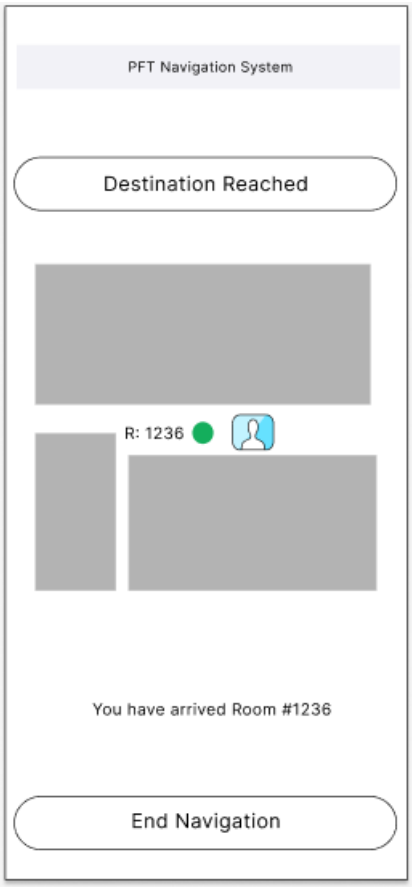

Figure 3: e. Destination Confirmation

Supported interaction:

- Confirm room location on the map
- End navigation

This screen helps reassure users that they have successfully reached the correct destination and allows them to end the navigation process easily.

3. Alternative Prototype Ideas

Before we decided on the final version of our PFT indoor navigation system, we explored a number of different low-fidelity prototype ideas. The final version of our prototype had a search-based interface that included an indoor map, a highlighted route, step-by-step instructions and a way to confirm the destination. But before we went that way, we looked at other design ideas that seemed good at first but turned out to be very flawed when we made low-fidelity prototypes.

One alternative idea was Augmented Reality (AR) navigation interface. In this idea, the user would hold their phone camera up and see virtual arrows or other visual cues on top of the real hallway to help them get to where they want to go. This idea sounded good because AR could make navigation feel more natural and intuitive by linking digital guidance directly to the real building space. By prototyping this idea, we found that users might benefit from help that feels immediate and is visually linked to their surroundings. It also helped us think more deeply about how important direction cues are at hallway intersections and other decision points. However, this idea did not make the final cut for several reasons. First, AR requires more advanced technical support, such as camera-based tracking, accurate positioning, and strong hardware performance. Second, it was difficult to represent meaningfully in a low-fidelity prototype because the main value of AR comes from dynamic, real-world interaction rather than static sketches or wireframes. Finally AR added complexity too early in the design process, while our project goal was to create a practical and understandable prototype that addressed core user needs first.

A second alternative idea was a text-only navigation interface. In this design users would receive step-by-step instructions without seeing a map. For example the system might display directions such as “Walk straight 30 meters,” “Turn right,” or “Destination on the left.” This approach helped us realize that simple instructions can reduce design complexity and are relatively easy to implement. It also showed that users often need direct action oriented guidance rather than too much visual information at once. From this prototype we learned that navigation support should be clear, brief and easy to follow, especially for users who are in a hurry. However this idea also revealed a major weakness. That is without visual map users may struggle to understand the spatial relationship between hallways, rooms, and building sections. This limitation is especially important in PFT, where corridors may look similar and building layouts can feel confusing. A text-only design may help at a basic level, but it does not provide the spatial awareness needed for confident indoor wayfinding. Because of that, it was not strong enough to support the full navigation experience we wanted to provide.

Through creating low-fidelity prototype we learned that users need both visual map and simple step-by-step instructions for indoor navigation. We learned that only giving text directions is not enough, because users may still feel confused if they do not understand the building layout. We also thought about ideas like AR navigation, which are interesting but too complicated for our project right now. We were able to figure out which ideas were useful and practical by making the low-fidelity prototype. In the end, it led us to a design that is simple, easy to use, and possible to make.

Participation (0.5 pages):

Sl	Participant Name	Email	Contribution Areas
1	Tania Khatun	tkhatu1@lsu.edu	Participated in brainstorming sessions, designed the low-fidelity prototype in Figma, and wrote the Low-Fidelity Prototype section
2	FNU Nurjahan	nurja1@lsu.edu	Participated in brainstorming sessions, designed the low-fidelity prototype in Figma, and contributed to writing the Low-Fidelity Prototype section
3	Nushrat Jahan Ria	nria1@lsu.edu	Participated in brainstorming sessions, designed the low-fidelity prototype in Figma, and contributed to writing the Low-Fidelity Prototype section
4	Sidhi Kumari Yadav	SidhiKumari.Yadav@lsu.edu	Participated in brainstorming sessions and wrote the Features section
5	Ming Li	Ming.Li@lsu.edu	Participated in brainstorming sessions and wrote the Features section